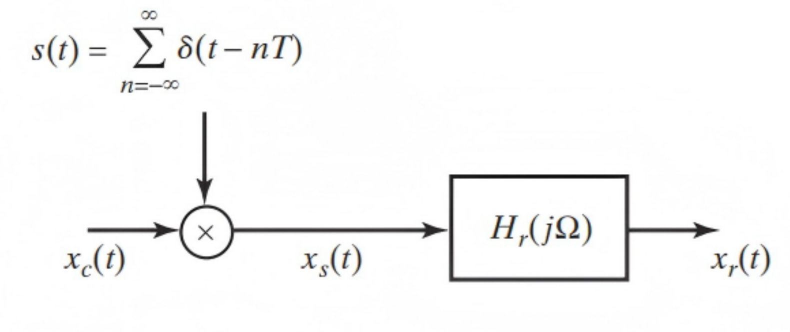
Problem Set 2

1 Consider the representation of the process of sampling followed by reconstruction shown in the following figure.



Assume that the input signal is

$$x_c(t) = 2\cos(100\pi t - \pi/4) + \cos(300\pi t + \pi/3), \qquad -\infty < t < \infty.$$

The frequency response of the reconstruction filter is

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| \le \pi/T, \\ 0, & |\Omega| > \pi/T. \end{cases}$$

(a) Determine the continuous-time Fourier transform $X_c(j\Omega)$ and plot it as a function of Ω.

(b) Assume that $f_s = 1/T = 500$ samples/sec and plot the Fourier transform $X_s(j\Omega)$ as a function of Ω for $-2\pi/T \le \Omega \le 2\pi/T$. What is the output $x_r(t)$ in this case? (You should be able to give an exact equation for $x_r(t)$.)

(c) Now, assume that $f_s = 1/T = 250$ samples/sec. Repeat part (b) for this condition.

2 Consider a stable discrete-time signal $x[n]$ whose discrete-time Fourier transform $X(e^{j\omega})$ satisfies the equation

$$X(e^{j\omega}) = X\left(e^{j(\omega-\pi)}\right)$$

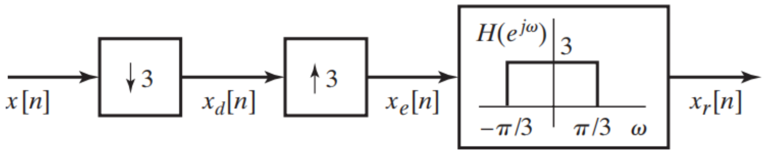
and has even symmetry, i.e., $x[n] = x[-n]$.

(a) Show that $X(e^{j\omega})$ is periodic with a period π.

(b) Find the value of $x[3]$. (*Hint:* Find values for all odd-indexed points.)

(c) Let $y[n]$ be the decimated version of $x[n]$, i.e., $y[n] = x[2n]$. Can you reconstruct $x[n]$ from $y[n]$ for all n? If yes, how? If no, justify your answer.

3 Consider the system shown in the following figure. For each of the following input signals $x[n]$, indicate whether the output $x_r[n] = x[n]$.



(a) $x[n] = \cos(\pi n/4)$
(b) $x[n] = \cos(\pi n/2)$
(c)

$$x[n] = \left[\frac{\sin(\pi n/8)}{\pi n}\right]^2$$

Hint: Use the modulation property of the Fourier transform to find $X(e^{j\omega})$.

4 Following are several z-transforms. For each, determine the inverse z-transform.

(a) $X(z) = \dfrac{1}{1 + \frac{1}{2}z^{-1}}, \qquad |z| > \dfrac{1}{2}.$

(b) $X(z) = \dfrac{1}{1 + \frac{1}{2}z^{-1}}, \qquad |z| < \dfrac{1}{2}.$

(c) $X(z) = \dfrac{1 - \frac{1}{2}z^{-1}}{1 + \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}, \qquad |z| > \dfrac{1}{2}.$

5 The input to a causal LTI system is

$$x[n] = u[-n-1] + \left(\frac{1}{2}\right)^n u[n].$$

The z-transform of the output of this system is

$$Y(z) = \frac{-\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + z^{-1}\right)}.$$

(a) Determine $H(z)$, the z-transform of the system impulse response. Be sure to specify the ROC.
(b) What is the ROC for $Y(z)$?
(c) Determine $y[n]$.